2.2.4 Unitary matrices ¶

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Homework 2.2.4.1. Let $Q \in \mathbb{C}^{m \times n}$ be an orthonormal matrix.

ALWAYS/SOMETIMES/NEVER: $Q^{-1} = Q^H$ and $QQ^H = I$.

▸ Answer ▸ Solution

If an orthonormal matrix is square, then it is called a unitary matrix.

Definition 2.2.4.1. Unitary matrix. Let $U \in \mathbb{C}^{m \times m}$. Then U is said to be a unitary matrix if and only if $U^H U = I$ (the identity).

Remark 2.2.4.2. Unitary matrices are always square. Sometimes the term ***orthogonal matrix*** is used instead of unitary matrix, especially if the matrix is real valued.

Unitary matrices have some very nice properties, as captured by the following exercises.

Homework 2.2.4.2. Let $Q \in \mathbb{C}^{m \times m}$ be a unitary matrix.

ALWAYS/SOMETIMES/NEVER: $Q^{-1} = Q^H$ and $QQ^H = I$.

▾ Answer ▾ Solution

If Q is unitary, then it is square and $Q^H Q = I$. Hence $Q^{-1} = Q^H$ and $QQ^H = I$.

ALWAYS

Now explain it!

Homework 2.2.4.3. TRUE/FALSE: If U is unitary, so is U^H.

▼ Answer ▼ Solution

Clearly, U^H is square. Also, $(U^H)^H U^H = (UU^H)^H = I$ by the last homework.

TRUE

Now prove it!

Homework 2.2.4.4. Let $U_0, U_1 \in \mathbb{C}^{m \times m}$ both be unitary.

ALWAYS/SOMETIMES/NEVER: $U_0 U_1$, is unitary.

▼ Answer ▼ Solution

Obviously, $U_0 U_1$ is a square matrix.

Now,

$$(U_0 U_1)^H (U_0 U_1) = U_1^H \underbrace{U_0^H U_0}_{I} U_1 = \underbrace{U_1^H U_1}_{I} = I.$$

Hence $U_0 U_1$ is unitary.

ALWAYS

Now prove it!

Homework 2.2.4.5. Let $U_0, U_1, \ldots, U_{k-1} \in \mathbb{C}^{m \times m}$ all be unitary.

ALWAYS/SOMETIMES/NEVER: Their product, $U_0 U_1 \cdots U_{k-1}$, is unitary.

▼ Answer ▼ Solution

Strictly speaking, we should do a proof by induction. But instead we will make the more informal argument that

$$
\begin{aligned}
(U_0 U_1 \cdots U_{k-1})^H U_0 U_1 \cdots U_{k-1} &= U_{k-1}^H \cdots U_1^H U_0^H U_0 U_1 \cdots U_{k-1} \\
&= \underbrace{U_{k-1}^H \underbrace{\cdots \underbrace{U_1^H \underbrace{U_0^H U_0}_{I} U_1}_{I} \cdots}_{I} U_{k-1}}_{I} = I.
\end{aligned}
$$

(When you see a proof that involed $\cdots$, it would be more rigorous to use a proof by induction.)

ALWAYS

Now prove it!

Remark 2.2.4.3. Many algorithms that we encounter in the future will involve the application of a sequence of unitary matrices, which is why the result in this last exercise is of great importance.

Perhaps the most important property of a unitary matrix is that it preserves length.

Homework 2.2.4.6. Let $U \in \mathbb{C}^{m \times m}$ be a unitary matrix and $x \in \mathbb{C}^m$. Prove that $\|Ux\|_2 = \|x\|_2$.

▼ Solution

$\|Ux\|_2^2$
$= \quad <$ alternative definition $>$
$(Ux)^H Ux$
$= \quad < (Az)^H = z^H A^H >$
$x^H U^H U x$
$= \quad < U$ is unitary $>$
$x^H x$
$= \quad <$ alternative definition $>$
$\|x\|_2^2.$

The converse is true as well:

Theorem 2.2.4.4. *If $A \in \mathbb{C}^{m \times m}$ preserves length ($\|Ax\|_2 = \|x\|_2$ for all $x \in \mathbb{C}^m$), then A is unitary.*

Proof.

We first prove that $(Ax)^H(Ay) = x^H y$ for all x, y by considering $\|x-y\|_2^2 = \|A(x-y)\|_2^2$. We then use that to evaluate $e_i^H A^H A e_j$.

Let $x, y \in \mathbb{C}^m$. Then

$\|x-y\|_2^2 = \|A(x-y)\|_2^2$
$\Leftrightarrow \quad <$ alternative definition $>$
$(x-y)^H(x-y) = (A(x-y))^H A(x-y)$
$= \quad < (Bz)^H = z^H B^H >$
$(x-y)^H(x-y) = (x-y)^H A^H A(x-y)$
$\Leftrightarrow \quad <$ multiply out $>$
$x^H x - x^H y - y^H x + y^H y = x^H A^H A x - x^H A^H A y - y^H A^H A x + y^H A^H A y$
$\Leftrightarrow \quad <$ alternative definition ; $\overline{x^H y} = y^H x >$
$\|x\|_2^2 - (x^H y + \overline{x^H y}) + \|y\|_2^2 = \|Ax\|_2^2 - (x^H A^H A y + \overline{x^H A^H A y}) + \|Ay\|_2^2$
$\Leftrightarrow \quad < \|Ax\|_2 = \|x\|_2$ and $\|Ay\|_2 = \|y\|_2; \alpha + \overline{\alpha} = 2\mathrm{Re}(\alpha) >$
$\mathrm{Re}\left(x^H y\right) = \mathrm{Re}\left((Ax)^H Ay\right)$

One can similarly show that $\mathrm{Im}\left(x^H y\right) = \mathrm{Im}\left((Ax)^H Ay\right)$ by considering $A(ix - y)$.

Conclude that $(Ax)^H(Ay) = x^H y$.

We now use this to show that $A^H A = I$ by using the fact that the standard basis vectors have the property that

$$e_i^H e_j = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{otherwise} \end{cases}$$

and that the i, j entry in $A^H A$ equals $e_i^H A^H A e_j$.

Note: I think the above can be made much more elegant by choosing α such that $\alpha x^H y$ is real and then looking at $\|x + \alpha y\|_2 = \|A(x + \alpha y)\|_2$ instead, much like we did in the proof of the Cauchy-Schwartz inequality. Try and see if you can work out the details.

Homework 2.2.4.7. Prove that if U is unitary then $\|U\|_2 = 1$.

▼ Solution

$$\begin{array}{l} \|U\|_2 \\ \quad = \quad < \text{definition} > \\ \max_{\|x\|_2=1} \|Ux\|_2 \\ \quad = \quad < \text{unitary matrices preserve length} > \\ \max_{\|x\|_2=1} \|x\|_2 \\ \quad = \quad < \text{algebra} > \\ 1 \end{array}$$

(The above can be really easily proven with the SVD. Let's point that out later.)

Homework 2.2.4.8. Prove that if U is unitary then $\kappa_2(U) = 1$.

▼ Solution

$$\begin{array}{l} \kappa_2 U \\ \quad = \quad < \text{definition} > \\ \|U\|_2 \|U^{-1}\|_2 \\ \quad = \quad < \text{both } U \text{ and } U^{-1} \text{ are unitary ; last homework} > \\ 1 \times 1 \\ \quad = \quad < \text{arithmetic} > \\ 1 \end{array}$$

The preservation of length extends to the preservation of norms that have a relation to the 2-norm:
d

Homework 2.2.4.9. Let $U \in \mathbb{C}^{m \times m}$ and $V \in \mathbb{C}^{n \times n}$ be unitary and $A \in \mathbb{C}^{m \times n}$. Show that

- $\|U^H A\|_2 = \|A\|_2$.
- $\|AV\|_2 = \|A\|_2$.

- $\|U^H AV\|_2 = \|A\|_2$.

▼ Hint ▼ Solution

- $$\begin{array}{l} \|U^H A\|_2 \\ \quad = \quad < \text{definition of 2-norm } > \\ \max_{\|x\|_2=1} \|U^H A x\|_2 \\ \quad = \quad < U \text{ is unitary and unitary matrices preserve length } > \\ \max_{\|x\|_2=1} \|A x\|_2 \\ \quad = \quad < \text{definition of 2-norm } > \\ \|A\|_2. \end{array}$$

- $$\begin{array}{l} \|AV\|_2 \\ \quad = \quad < \text{definition of 2-norm } > \\ \max_{\|x\|_2=1} \|AVx\|_2 \\ \quad = \quad < V^H \text{ is unitary and unitary matrices preserve length } > \\ \max_{\|Vx\|_2=1} \|A(Vx)\|_2 \\ \quad = \quad < \text{substitute } y = Vx > \\ \max_{\|y\|_2=1} \|Ay\|_2 \\ \quad = \quad < \text{definition of 2-norm } > \\ \|A\|_2. \end{array}$$

- The last part follows immediately from the previous two:

$$\|U^H AV\|_2 = \|U^H (AV)\|_2 = \|AV\|_2 = \|A\|_2.$$

Exploit the definition of the 2-norm:

$$\|A\|_2 = \max_{\|x\|_2=1} \|Ax\|_2.$$

Homework 2.2.4.10. Let $U \in \mathbb{C}^{m\times m}$ and $V \in \mathbb{C}^{n\times n}$ be unitary and $A \in \mathbb{C}^{m\times n}$. Show that

- $\|U^H A\|_F = \|A\|_F$.
- $\|AV\|_F = \|A\|_F$.
- $\|U^H AV\|_F = \|A\|_F$.

▼ Hint ▼ Solution

- Partition

$$A = \left(\begin{array}{c|c|c} a_0 & \cdots & a_{n-1} \end{array} \right).$$

Then we saw in Subsection 1.3.3 that $\|A\|_F^2 = \sum_{j=0}^{n-1} \|a\|_2^2$.

Now,

$$
\begin{array}{l}
\|U^H A\|_F^2 \\
\quad = \quad < \text{partition } A \text{ by columns} > \\
\|U^H \left(\begin{array}{c|c|c} a_0 & \cdots & a_{n-1} \end{array} \right) \|_F^2 \\
\quad = \quad < \text{property of matrix-vector multiplication} > \\
\| \left(\begin{array}{c|c|c} U^H a_0 & \cdots & U^H a_{n-1} \end{array} \right) \|_F^2 \\
\quad = \quad < \text{exercise in Chapter 1} > \\
\sum_{j=0}^{n-1} \|U^H a_j\|_2^2 \\
\quad = \quad < \text{unitary matrices preserve length} > \\
\sum_{j=0}^{n-1} \|a_j\|_2^2 \\
\quad = \quad < \text{exercise in Chapter 1} > \\
\|A\|_F^2.
\end{array}
$$

- To prove that $\|AV\|_F = \|A\|_F$ recall that $\|A^H\|_F = \|A\|_F$.
- The last part follows immediately from the first two parts.

How does $\|A\|_F$ relate to the 2-norms of its columns?

In the last two exercises we consider $U^H AV$ rather than UAV because it sets us up better for future discussion.